

Q1 Let x and Y be two continuous random variables have following joint pdf

$$f(x, y) = Axy ; 0 < x < 1 ; 0 < y < 1$$

- i) Find 'A' constant.
- ii) Find the marginal density function of x and Y .
- iii) Check whether x and Y are independent or not.

sol Here,

$$f(x, y) = Axy ; 0 < x < 1 ; 0 < y < 1$$

i) By the definition of joint pdf,

$$\int_x \int_y f(x, y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^1 Axy dy dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{y^2}{2} \right]_0^1 dx = 1$$

$$\Rightarrow A \int_0^1 x \cdot \frac{1}{2} dx = 1$$

$$\Rightarrow \frac{A}{2} \left[\frac{x^2}{2} \right]_0^1 = 1$$

$$\Rightarrow \frac{A}{2} \cdot \frac{1}{2} = 1$$

$$\therefore A = 4$$

Hence, the constant $A = 4$ Ans

$$\therefore f(x, y) = 4xy ; 0 < x < 1, 0 < y < 1.$$

ii) $f(x, y) = 4xy$; $0 < x < 1$; $0 < y < 1$

The marginal pdf of x is given by

$$f(x) = \int_y f(x, y) dy = \int_0^1 4xy dy = 4x \left[\frac{y^2}{2} \right]_0^1 = 2x \text{ Ans}$$

The marginal pdf of y is given by

$$f(y) = \int_x f(x, y) dx = \int_0^1 4xy dx = 4y \left[\frac{x^2}{2} \right]_0^1 = 2y \text{ Ans}$$

iii) Here,

$$f(x, y) = 4xy$$

$$f(x) = 2x$$

$$f(y) = 2y$$

$$\therefore f(x) \cdot f(y) = 4xy$$

$$\therefore f(x, y) = f(x) \cdot f(y)$$

Thus, the random variables x and y are independent.

Ans

2) Let x and Y be the continuous random variables having following joint pdf.

$$f(x, y) = Axy; \quad 0 < x < 1; \quad 0 < y < x$$

i) Find the value of A .

ii) Find the marginal pdf of x and Y .

iii) Check whether x and Y are independent or not.

Here

$$f(x, y) = Axy; \quad 0 < x < 1, \quad 0 < y < x$$

i) By the definition of joint pdf,

$$\iint f(x, y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^x Axy dy dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = 1$$

$$\Rightarrow A \int_0^1 x \cdot \frac{x^2}{2} dx = 1$$

$$\Rightarrow \frac{A}{2} \int_0^1 x^3 dx = 1$$

$$\Rightarrow \frac{A}{2} \cdot \left[\frac{x^4}{4} \right]_0^1 = 1$$

$$\Rightarrow \frac{A}{2} \cdot \frac{1}{4} = 1$$

$$\therefore A = 8$$

Hence, the value of A is 8 Ans

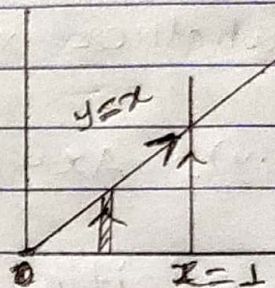
ii) Here,

$$f(x, y) = 8xy; 0 < x < 1; 0 < y < x$$

The marginal pdf of x is given by

$$\begin{aligned} f(x) &= \int_y^x 8xy \, dy = \int_0^x 8xy \, dy \\ &= 8x \cdot \left[\frac{y^2}{2} \right]_0^x = 8x \cdot \frac{x^2}{2} \end{aligned}$$

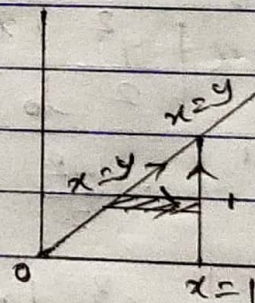
$$\therefore f(x) = 4x^3 \quad \underline{\text{Ans}}$$



The marginal pdf of y is given by

$$\begin{aligned} f(y) &= \int_x^1 8xy \, dx = \int_y^1 8xy \, dx \\ &= 8y \cdot \left[\frac{x^2}{2} \right]_y^1 = 8y \cdot \left(\frac{1}{2} - \frac{y^2}{2} \right) \end{aligned}$$

$$\therefore f(y) = 4y(1-y^2) \quad \underline{\text{Ans}}$$



iii) Here,

$$\begin{aligned} f(x) \cdot f(y) &= 4x^3 \cdot 4y(1-y^2) = 16x^3y(1-y^2) \\ f(x, y) &= 8xy \end{aligned}$$

$$\therefore f(x, y) \neq f(x) \cdot f(y). \quad \text{so}$$

Thus, x and y are not independent. Ans

Conditional Type Questions

Let X and Y be two dimensional random variables having joint pdf

$$f(x, y) = Axy ; 0 < x < 2, 0 < y < 1.$$

Find the value of A .

Find the marginal pdf of X and Y .

Check X and Y are independent or not.

Find the conditional pdf of X given Y .

Find the conditional pdf of Y given X .

Here,

$$f(x, y) = Axy ; 0 < x < 2, 0 < y < 1$$

By the definition of joint pdf,

$$\int_0^2 \int_0^1 f(x, y) dy dx = 1$$

$$\Rightarrow \int_0^2 \int_0^1 Axy dy dx = 1$$

$$\Rightarrow A \int_0^2 x \left[\frac{y^2}{2} \right]_0^1 dx = 1$$

$$\Rightarrow A \int_0^2 x \cdot \frac{1}{2} dx = 1$$

$$\Rightarrow \frac{A}{2} \int_0^2 x dx = 1$$

$$\Rightarrow 2A \cdot \left[\frac{x^2}{2} \right]_0^2 = 1$$

$$\Rightarrow \frac{A}{2} \cdot \frac{4}{2} = 1$$

$$\therefore A = 1$$

Hence, the required value of A is 1.

ii) Here,

$$f(x, y) = xy \quad ; \quad 0 < x < 2; 0 < y < 1$$

The marginal pdf of x is given by

$$f(x) = \int_y xy \, dy = \int_0^1 xy \, dy = x \left[\frac{y^2}{2} \right]_0^1 = \frac{x}{2} \quad \underline{\text{Ans}}$$

The marginal pdf of y is given by

$$f(y) = \int_x xy \, dx = \int_0^2 xy \, dx = y \left[\frac{x^2}{2} \right]_0^2 = y \cdot \frac{4}{2} = 2y \quad \underline{\text{Ans}}$$

iii) Here,

$$f(x) = x/2$$

$$f(y) = 2y$$

$$\therefore f(x) \cdot f(y) = \frac{x}{2} \cdot 2y = xy$$

$$\therefore f(x, y) = f(x) \cdot f(y) = xy$$

Hence, x and y are independent.

iv) The conditional pdf of x given y is:

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{xy}{2y} = \frac{x}{2} \quad \underline{\text{Ans}}$$

3) The conditional pdf of Y given x :

$$f(Y/x) = \frac{f(x,y)}{f(x)} = \frac{xy}{x/2} = 2y \quad \underline{\text{Ans}}$$

2022-Fall, Pre-Board, NAST

4) The joint pdf of two random variables x and Y is given below:

$$f(x,y) = K(6-x-y) \quad ; \quad 0 < x < 2, \quad 2 < y < 4$$

i) Find the value of constant K .

ii) Find the marginal pdf of x and Y .

iii) Find the $E(x)$ and $V(x)$.

iv) Are x and Y are independent random variables?

v) Find the conditional pdf of x and Y .

eg Here,

$$f(x,y) = K(6-x-y) \quad ; \quad 0 < x < 2, \quad 2 < y < 4$$

i) By the definition of joint pdf,

$$\int_0^2 \int_2^4 f(x,y) dy dx = 1$$

$$\Rightarrow \int_0^2 \int_2^4 K(6-x-y) dy dx = 1$$

$$\Rightarrow K \int_0^2 \left[6y - xy - \frac{y^2}{2} \right]_2^4 dx = 1$$

$$\Rightarrow K \int_0^2 \left(6 \times 4 - 4x - \frac{16}{2} = 12 + 2x + \frac{4}{2} \right) dx = 1$$

$$\Rightarrow K \int_0^2 (6 - 2x) dx = 1$$

$$\Rightarrow K \left[6x - \frac{2x^2}{2} \right]_0^2 = 1$$

$$\Rightarrow K (6 \times 2 - 2 \times \frac{2^2}{2}) = 1$$

$$\Rightarrow K (12 - 4) = 1$$

$$\Rightarrow 8K = 1$$

$$\therefore K = \frac{1}{8}$$

Hence, the value of constant K is $\frac{1}{8}$. Ans

ii) Here,

$$f(x, y) = \frac{1}{8} (6 - x - y) ; 0 < x < 2 ; 2 < y < 4$$

The marginal pdf of x is given by

$$f(x) = \int_y f(x, y) dy = \int_2^4 \frac{1}{8} (6 - x - y) dy$$

$$= \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4 = \frac{1}{8} \left[6 \times 4 - 4x - \frac{4^2}{2} - \left(6 \times 2 + 2x + \frac{2^2}{2} \right) \right]$$

$$\therefore f(x) = \frac{1}{8} (6 - 2x) = \frac{1}{4} (3 - x) \quad \underline{\underline{\text{Ans}}}$$

The marginal pdf of X is given by

$$f(y) = \int_x f(x,y) dx = \int_0^2 \frac{1}{8} (6-x-y) dx$$

$$= \frac{1}{8} \left[6x - \frac{x^2}{2} - xy \right]_0^2 = \frac{1}{8} (6 \times 2 - \frac{2^2}{2} - 2y)$$

$$f(y) = \frac{1}{8} (10-2y) = \frac{1}{4} (5-y) \quad \underline{\text{Ans}}$$

$$\text{iii) } E(X) = \int_x x f(x,y) dx = \int_0^2 \frac{x}{8} (6-x-y) dx = \int_0^2 \frac{1}{8} (6x - x^2 - xy) dx$$

$$= \frac{1}{8} \left[\frac{6x^2}{2} - \frac{x^3}{3} - \frac{x^2 y}{2} \right]_0^2 = \frac{1}{8} \left(\frac{6 \times 2^2}{2} - \frac{2^3}{3} - \frac{2^2 \cdot y}{2} \right)$$

$$\therefore E(X) = \frac{1}{8} \left(\frac{24}{1} - \frac{8}{3} - 2y \right) \quad \underline{\text{Ans}}$$

Also,

$$E(X^2) = \int_x x^2 f(x,y) dx = \int_0^2 \frac{x^2}{8} (6-x-y) dx = \frac{1}{8} \int_0^2 (6x^2 - x^3 - x^2 y) dx$$

$$= \frac{1}{8} \left[\frac{6x^3}{3} - \frac{x^4}{4} + \frac{x^3 y}{3} \right]_0^2 = \frac{1}{8} \left[\frac{6 \times 2^3}{3} - \frac{2^4}{4} + \frac{2^3 y}{3} \right]$$

$$\therefore E(X^2) = \frac{1}{8} (12 - \frac{8}{1} + \frac{8y}{3}) = \frac{1}{2} \left(3 - \frac{2y}{3} \right)$$

$$i) \therefore V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{1}{2} \left(3 - \frac{2y}{3} \right) - \left[\frac{1}{8} \left(\frac{28}{3} - 2y \right) \right]^2$$

$$= \frac{3}{2} - \frac{y}{3} - \frac{1}{8^2} \left(\left(\frac{28}{3} \right)^2 - 2 \cdot \frac{28}{3} \cdot 2y + (2y)^2 \right)$$

$$= \frac{3}{2} - \frac{y}{3} - \frac{1}{64} \left(\frac{784}{9} - \frac{112}{3} y + 4y^2 \right)$$

$$= \frac{3}{2} - \frac{y}{3} - \frac{49}{36} + \frac{7y}{12} - \frac{1}{16} y^2$$

$$\therefore V(x) = \frac{5}{36} + \frac{1}{4} y - \frac{1}{16} y^2$$

$$\therefore V(x) = -\frac{1}{16} y^2 + \frac{1}{4} y + \frac{5}{36} \quad \underline{\text{Ans}}$$

iv) Here,

$$f(x) = \frac{1}{4} (3-x)$$

$$f(y) = \frac{1}{4} (5-y)$$

$$\therefore f(x) \cdot f(y) = \frac{1}{4} (3-x) \cdot \frac{1}{4} (5-y) = \frac{1}{16} (3-x)(5-y)$$

$$\therefore f(x, y) \neq f(x) \cdot f(y)$$

Hence, x and y are not independent.

∴ The conditional pdf of x given Y is:

$$f(x/y) = \frac{f(x,y)}{f(y)} = \frac{\frac{1}{8} \cdot (6-x-y)}{\frac{1}{4} (5-y)} = \frac{1}{2} \cdot \frac{(6-x-y)}{(5-y)} \quad \underline{\underline{Ans}}$$

The conditional pdf of Y given x is:

$$f(y/x) = \frac{f(x,y)}{f(x)} = \frac{\frac{1}{8} \cdot (6-x-y)}{\frac{1}{4} (3-x)} = \frac{(6-x-y)}{2(3-x)} \quad \underline{\underline{Ans}}$$

2023-fall, Pre-Board (NAST)

5. 4a) If $f(x,y) = Axy, 0 < x < 1, 0 < y < 2$
 $= 0, \text{ otherwise}$

i) Find the value of A.

ii) Find the marginal distribution of X and Y.

iii) Find the conditional distribution of X and Y.

iv) Find whether X and Y are independent or not.

~~So~~ Here,

$$f(x,y) = Axy, 0 < x < 1, 0 < y < 2$$
$$= 0, \text{ otherwise}$$

i) By the definition of joint pdf,

$$\int_0^1 \int_0^2 f(x,y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^2 Axy dy dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{y^2}{2} \right]_0^2 dx = 1$$

$$\Rightarrow A \int_0^1 x \cdot \left(\frac{2^2}{2} \right) dx = 1$$

$$\Rightarrow 2A \int_0^1 x dx = 1$$

$$\Rightarrow 2A \left[\frac{x^2}{2} \right]_0^1 = 1$$

$$\Rightarrow 2A \cdot \frac{1^2}{2} = 1$$

$$\therefore A = 1$$

Hence, the value of A is 1. Ans

ii) Here,
 $f(x, y) = xy; 0 < x < 1, 0 < y < 2$

The marginal distribution of x and y is given by

$$f(x) = \int_x f(x, y) dy = \int_0^2 xy dy = x \left[\frac{y^2}{2} \right]_0^2 = x \cdot \frac{2^2}{2} = 2x \quad \underline{\underline{\text{Ans}}}$$

The marginal distribution of y and x is given by

$$f(y) = \int_x f(x, y) dx = \int_0^1 xy dx = y \left[\frac{x^2}{2} \right]_0^1 = y \cdot \frac{1^2}{2} = \frac{y}{2} \quad \underline{\underline{\text{Ans}}}$$

iii) The conditional distribution of x is given by

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{xy}{y/2} = 2x \quad \underline{\underline{\text{Ans}}}$$

The conditional distribution of y is given by

$$f(y/x) = \frac{f(x, y)}{f(x)} = \frac{xy}{2x} = \frac{y}{2} \quad \underline{\underline{\text{Ans}}}$$

(iv) Here,

$$f(x) = 2x$$

$$f(y) = y/2$$

$$\therefore f(x) \cdot f(y) = 2x \cdot \frac{y}{2} = xy$$

$$\therefore f(xy) = f(x) \cdot f(y) = xy$$

Hence, X and Y are independent. Ans

2023-Spring

6) 4.b) Two random variables x and y having the following probability density function.

$$f(x, y) = 8xy, \quad 0 < x < y < 1$$
$$0, \quad \text{otherwise}$$

find:

- i) Marginal probability density function of x and y .
- ii) Conditional functions of x on y and y on x .

Here,

$$f(x, y) = 8xy, \quad 0 < x < y < 1$$
$$= 0, \quad \text{otherwise}$$

i) The marginal pdf of x is given by

$$f(x) = \int_y f(x, y) dy = \int_x^1 8xy dy$$
$$= 8x \cdot \left[\frac{y^2}{2} \right]_x^1 = 8x \left(\frac{1}{2} - \frac{x^2}{2} \right)$$

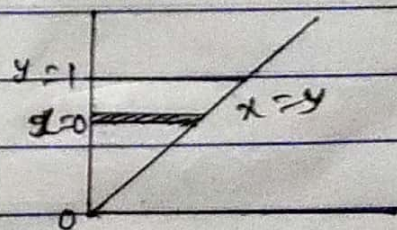
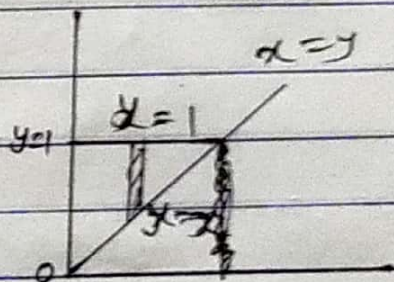
$$\therefore f(x) = \frac{8x}{2} (1 - x^2) = 4x(1 - x^2) \text{ Ans}$$

Similarly,

The marginal pdf of y is given by

$$f(y) = \int_x f(x, y) dx = \int_0^y 8xy dx$$
$$= 8y \left[\frac{x^2}{2} \right]_0^y = 8y \cdot \frac{y^2}{2}$$

$$\therefore f(y) = 4y^3 \text{ Ans}$$



(c) ii)

The conditional function of X on Y is given by

$$f(X/Y) = \frac{f(x,y)}{f(y)} = \frac{8xy}{4y^3} = \frac{2x}{y^2} \quad \underline{\underline{Ans}}$$

The condition function of Y on X is given by

$$f(Y/X) = \frac{f(x,y)}{f(x)} = \frac{8xy}{4x(1-x^2)} = \frac{2y}{1-x^2} \quad \underline{\underline{Ans}}$$

2022-Fall

python, NAST

4b) Let X and Y be two continuous random variables with joint pdf,

$$f(x, y) = Axy \quad ; \quad 0 < x < 1 \text{ and } 0 < y < x$$

i) Find the constant A .

ii) Find the marginal density function of X and Y .

iii) Find the conditional density function of Y given $X=x$.

iv) Check the independency of X and Y .

Here,

$$f(x, y) = Axy \quad ; \quad 0 < x < 1 \text{ \& } 0 < y < x$$

i) By the definition of joint pdf,

$$\iint_{xy} f(x, y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^x (Axy) dy dx = 1$$

$$\Rightarrow A \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = 1$$

$$\Rightarrow A \int_0^1 x \frac{x^2}{2} dx = 1$$

$$\Rightarrow \frac{A}{2} \int_0^1 x^3 dx = 1$$

$$\Rightarrow \frac{A}{2} \left[\frac{x^4}{4} \right]_0^1 = 1$$

$$\Rightarrow \frac{A}{2} \cdot \frac{1}{4} = 1$$

$$\Rightarrow A = 8$$

$\therefore$ The required constant A is 8. Ans

ii) Here,

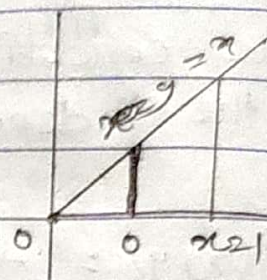
$$f(x, y) = 8xy \quad ; \quad 0 < x < 1 \text{ and } 0 < y < x$$

The marginal density function of x is given by

$$f(x) = \int_y f(x, y) dy = \int_0^x 8xy dy$$

$$= 8x \cdot \left[\frac{y^2}{2} \right]_0^x = 8x \cdot \frac{x^2}{2}$$

$$\therefore f(x) = 4x^3 \quad \underline{\underline{Ans}}$$

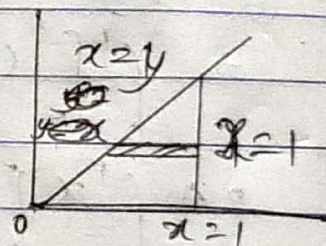


The marginal density function of y is given by

$$f(y) = \int_x f(x, y) dx = \int_y^1 8xy dx$$

$$= 8y \cdot \left[\frac{x^2}{2} \right]_y^1 = 8y \cdot \left(\frac{1}{2} - \frac{y^2}{2} \right)$$

$$\therefore f(y) = 4y(1-y^2) \quad \underline{\underline{Ans}}$$



iii) The conditional density function of y given $x=x$ is,

$$f(y/x) = \frac{f(x, y)}{f(x)} = \frac{8xy}{4x^3} = \frac{2y}{x^2} \quad \underline{\underline{Ans}}$$

The conditional density function of x given $y=y$ is,

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{8xy}{4y(1-y^2)} = \frac{2x}{1-y^2} \quad \underline{\underline{Ans}}$$

iv) Here,

$$f(x)f(y) = 4x^3 \cdot 4y(1-y^2) = 16x^3 y(1-y^2)$$

$$\therefore f(x,y) \neq f(x) \cdot f(y).$$

Thus, x and y are not independent. Ans

2021-Fall

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4.b) If two random variables have the joint pdf
 $f(x,y) = \frac{2}{3}(x+2y)$; $0 < x < 1$ and $0 < y < 1$

- i) Find the marginal pdf of x and y .
- ii) Find the conditional density of x given that $Y=y$.
- iii) Are X and Y are independent?

solⁿ Here,

$$f(x,y) = \frac{2}{3}(x+2y); \quad 0 < x < 1, 0 < y < 1$$

i) Here,

The marginal pdf of x is given by

$$f(x) = \int_y f(x,y) dy = \int_0^1 \frac{2}{3}(x+2y) dy$$

$$= \frac{2}{3} \left[xy + \frac{2y^2}{2} \right]_0^1 = \frac{2}{3} \left(x + \frac{2x^2}{2} \right) = \frac{2}{3}(x+1) \quad \underline{\underline{Ans}}$$

The marginal pdf of y is given by

$$f(y) = \int_x f(x,y) dx = \int_0^1 \frac{2}{3}(x+2y) dx = \left[\frac{2}{3} \left(\frac{x^2}{2} + 2xy \right) \right]_0^1$$
$$= \frac{2}{3} \left(\frac{1}{2} + 2y \right) = \frac{2}{3} \left(\frac{1}{2} + 2y \right) \quad \underline{\underline{Ans}}$$

ii) The conditional density of X given $Y=y$ is:

$$\begin{aligned} f(X/Y=y) &= \frac{f(x, y)}{f(y)} = \frac{\frac{2}{3}(x+2y)}{\frac{2}{3}(\frac{1}{2}+2y)} \\ &= \frac{x+2y}{\frac{1}{2}+2y} \quad \underline{\text{Ans}} \end{aligned}$$

iii) Here,

$$\begin{aligned} f(x) \cdot f(y) &= \frac{2}{3}(x+1) \cdot \frac{2}{3}\left(\frac{1}{2}+2y\right) \\ &\neq f(x, y) \end{aligned}$$

Hence, X and Y are not independent. Ans

2020-Fall / 2018-spring

(9)

4.6) Joint probability distribution of x and y is given

$$f(x, y) = 4xy e^{-(x^2+y^2)} ; x > 0 ; y \geq 0$$

i) Test whether x and y are independent or not?

ii) Find the conditional density of x given $Y=y$.

sol Here,

$$f(x, y) = 4xy e^{-(x^2+y^2)} ; x > 0, y \geq 0$$

i) To check independency, we have to calculate marginal pdf.

The marginal pdf of x and y are:

$$f(x) = \int_y f(x, y) dy = \int_0^{\infty} 4xy e^{-(x^2+y^2)} dy$$

$$= 4x \int_0^{\infty} y e^{-(x^2+y^2)} dy$$

$$\Rightarrow f(x) = \int_0^{\infty} 4xy e^{-x^2} \cdot e^{-y^2} dy = 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy$$

$$\text{put } y^2 = t$$

$$\therefore 2y dy = dt$$

$$\text{when } y=0, t=0$$

$$\text{when } y=\infty, t=\infty$$

$$\begin{aligned} \Rightarrow f(x) &= 4x e^{-x^2} \int_0^{\infty} e^{-t} \frac{dt}{2} = \frac{4x e^{-x^2}}{2} \left[\frac{e^{-t}}{-1} \right]_0^{\infty} \\ &= \frac{4x e^{-x^2}}{2} (0 - 1) = -2x e^{-x^2} \end{aligned}$$

$$\therefore f(x) = 2xe^{-x^2} = 2xe^{-x^2}$$

Also,

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} f(x, y) dx = \int_0^{\infty} 4xy e^{-(x^2+y^2)} dx$$

$$\Rightarrow f(y) = 4ye^{-y^2} \int_0^{\infty} xe^{-x^2} dx$$

$$\text{put } x^2 = u$$

$$\therefore x^2 = u$$

$$\Rightarrow 2x dx = du$$

$$\text{when } x=0, u=0$$

$$\text{when } x=\infty, u=\infty$$

$$\therefore f(y) = 4ye^{-y^2} \int_0^{\infty} \frac{e^{-u} du}{2} = 2ye^{-y^2} \int_0^{\infty} e^{-u} du$$

$$\Rightarrow f(y) = 2ye^{-y^2} \left[\frac{e^{-u}}{-1} \right]_0^{\infty} = 2ye^{-y^2} (0 - 1)$$

$$\therefore f(y) = 2ye^{-y^2}$$

Now,

$$f(x) \cdot f(y) = 2xe^{-x^2} \cdot 2ye^{-y^2} = 4xy e^{-(x^2+y^2)}$$

$$\therefore f(x, y) = f(x) \cdot f(y) = 2xe^{-(x^2+y^2)}$$

Hence, X and Y are independent. Ans

(ii) The conditional density of X given $Y=y$ is ;

$$\begin{aligned} f(X/Y) &= \frac{f(x,y)}{f(y)} = \frac{4xy e^{-(x^2+y^2)}}{2ye^{-y^2}} \\ &= 2x e^{-x^2} \quad \text{Ans} \end{aligned}$$

from Book

10. If $f(x, y) = e^{-(x+y)}$, $x \geq 0, y \geq 0$
 $= 0$, otherwise

is the joint pdf of x given in random variables
 x and y .

find

i) $P(x < 1)$

ii) the marginal probability density of x and y .

iii) examine whether x and y are independent or not.

Here,

$$f(x, y) = e^{-(x+y)}, \quad x \geq 0, y \geq 0$$

$$i) P(x < 1) = \int_0^1 \int_0^{\infty} e^{-(x+y)} dx dy$$

$$= \int_0^1 e^{-x} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} dy$$

$$= \int_0^1 e^{-x} \left(\frac{0-1}{-1} \right) dy$$

$$= \int_0^1 e^{-x} dy$$

$$= \left[\frac{e^{-x}}{-1} \right]_0^1$$

$$= \frac{(e^{-1} - 1)}{-1} = -e^{-1} + 1$$

$$\therefore P(x < 1) = 1 - e^{-1}$$

(ii) The marginal pdf of x is given by

$$\begin{aligned} f(x) &= \int_y f(x,y) dy = \int_0^{\infty} e^{-(x+y)} dy \\ &= e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x} \left[\frac{e^{-y}}{-1} \right]_0^{\infty} \\ &= e^{-x} \left(\frac{e^{-\infty} - e^0}{-1} \right) = e^{-x} \left(\frac{0 - 1}{-1} \right) \end{aligned}$$

$$\therefore f(x) = e^{-x} \quad \underline{\text{Ans}}$$

The marginal pdf of y is given by

$$\begin{aligned} f(y) &= \int_x f(x,y) dx = \int_0^{\infty} e^{-(x+y)} dx \\ &= e^{-y} \int_0^{\infty} e^{-x} dx = e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^{\infty} \\ &= e^{-y} \left(\frac{e^{-\infty} - e^0}{-1} \right) = e^{-y} \left(\frac{0 - 1}{-1} \right) \end{aligned}$$

$$\therefore f(y) = e^{-y} \quad \underline{\text{Ans}}$$

iii) Here,

$$f(x) \cdot f(y) = e^{-x} \cdot e^{-y} = e^{-(x+y)}$$

$$\therefore f(x,y) = f(x) \cdot f(y) = e^{-(x+y)}$$

Hence, x and y are independent.

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prob

11) Suppose that the joint pdf of random variables X and Y is given by probability function

$$f(x, y) = 4xy, \quad 0 < x < 1, \quad 0 < y < 1 \\ = 0, \quad \text{otherwise}$$

- verify that the joint pdf probability is a joint pdf.
- Find $P(X < 0.5)$ and $P(Y < 0.8)$.
- Find the marginal distribution of X and Y .
- Find the conditional distribution of X given Y .
- Is X and Y are independent?
- Find $E(X)$, $E(Y)$, $V(X)$ and $V(Y)$.
- Find the covariance between X and Y .

11) i) $\int_0^1 \int_0^1 4xy \, dy \, dx = 4 \int_0^1 x \left[\frac{y^2}{2} \right]_0^1 dx = \frac{4}{2} \int_0^1 x \cdot 1 \, dx$
 $= 2 \left[\frac{x^2}{2} \right]_0^1 = \frac{2}{2} \cdot 1^2 = 1$

ii) $f(x, y) \geq 0$ for $0 < x < 1, 0 < y < 1$

iii) $\int_0^{0.25} \int_0^{0.5} 4xy \, dy \, dx = 4 \left[\frac{x^2}{2} \right]_0^{0.25} \left[\frac{y^2}{2} \right]_0^{0.5} = 4 \times \frac{0.0625}{2} \times \frac{0.25}{2}$
 $= 0.015625$

which is well defined.

$\therefore f(x, y)$ is joint pdf.

b) $P(X < 0.5 \text{ and } Y < 0.8)$

$$= \int_0^{0.5} \int_0^{0.8} 4xy \, dy \, dx = 4 \left[\frac{x^2}{2} \right]_0^{0.5} \left[\frac{y^2}{2} \right]_0^{0.8}$$

$$= 4 \left(\frac{0.5^2}{2} \times \frac{0.8^2}{2} \right) = 0.16 \text{ Ans}$$

c) The marginal pdf of x and y is given by:

$$f(x) = \int_y f(x,y) \, dy = \int_0^1 4xy \, dy = 4x \left[\frac{y^2}{2} \right]_0^1 = 2x \text{ Ans}$$

$$f(y) = \int_x f(x,y) \, dx = \int_0^1 4xy \, dx = 4y \left[\frac{x^2}{2} \right]_0^1 = 2y \text{ Ans}$$

d) The conditional distribution of x given y is:

$$f(x/y) = \frac{f(x,y)}{f(y)} = \frac{4xy}{2y} = 2x \text{ Ans}$$

e) Here,

$$f(x) \cdot f(y) = 2x \cdot 2y = 4xy$$

$$\therefore f(x,y) = f(x) \cdot f(y) = 4xy$$

Hence, x and y are independent. Ans

$$E(X) = \int_x x f(x, y) dx = \int_0^1 x \cdot 4xy dx = \int_0^1 4x^2 y dx$$

$$= 4y \left[\frac{x^3}{3} \right]_0^1 = 4y \cdot \frac{1}{3} = \frac{4y}{3}$$

$$E(X) = \int_x x f(x) dx = \int_0^1 x \cdot 2x dx = \int_0^1 2x^2 dx$$

$$= \left[\frac{2x^3}{3} \right]_0^1 = \frac{2}{3} \text{ Ans}$$

$$E(X^2) = \int_x x^2 f(x) dx = \int_0^1 x^2 \cdot 2x dx = \int_0^1 2x^3 dx$$

$$= \left[\frac{2x^4}{4} \right]_0^1 = \frac{2 \cdot 1}{4} = \frac{1}{2}$$

$$\therefore V(X) = E(X^2) - [E(X)]^2$$

$$= \frac{1}{2} - \left(\frac{2}{3} \right)^2 = \frac{1}{18} \text{ Ans}$$

$$E(Y) = \int_y y f(y) dy = \int_0^1 y \cdot 2y dy = \int_0^1 2y^2 dy$$

$$= \left[\frac{2y^3}{3} \right]_0^1 = \frac{2}{3}$$

$$\therefore E(Y^2) = \int_y y^2 f(y) dy = \int_0^1 y^2 \cdot 2y dy = 2 \left[\frac{y^4}{4} \right]_0^1$$

$$= \frac{2}{4} = \frac{1}{2}$$

$$\begin{aligned}\therefore V(Y) &= E(Y^2) - [E(Y)]^2 \\ &= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \\ &= \frac{1}{18} \text{ Ans}\end{aligned}$$

$$Q \rightarrow \text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$\begin{aligned}\text{where, } E(XY) &= \int_0^1 \int_0^1 xy f(x, y) dy dx = \int_0^1 \int_0^1 xy \cdot 4xy dy dx \\ &= \int_0^1 \int_0^1 4x^2 y^2 dy dx = 4 \left[\frac{x^3}{3} \right]_0^1 \left[\frac{y^3}{3} \right]_0^1 \\ &= \frac{4}{9}\end{aligned}$$

$$\therefore E(X) \cdot E(Y) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

$$\therefore \text{Cov}(X, Y) = \frac{4}{9} - \frac{4}{9} = 0 \text{ Ans}$$

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4.6) Suppose X and Y have the following joint pmf.

$Y \backslash X$	2	3
1	0.1	0.15
2	0.2	0.3
3	0.1	0.15

Find:

- The marginal distribution of X and Y
- The conditional probability of Y given $X = 3$.
- $F(2, 2)$ and $F(3, 2)$.
- Variance of X and variance of Y .

For marginal distribution:

$Y \backslash X$	2	3	Total
1	0.1	0.15	0.25
2	0.2	0.3	0.5
3	0.1	0.15	0.25
Total	0.4	0.6	1

i) The marginal distribution of X is:

~~$P_X(2) = 0.4$~~ $P(X=2) = 0.4$ Ans

$P(X=3) = 0.6$ Ans

$P(Y=1) = 0.25$ Ans

$P(Y=2) = 0.5$ Ans

$P(Y=3) = 0.25$ Ans

ii) The conditional probability of Y given that $X=3$ is;

$$P(Y=1/X=3) = \frac{P(3,1)}{P(3)} = \frac{0.15}{0.6} = 0.25 \text{ Ans}$$

$$P(Y=2/X=3) = \frac{P(3,2)}{P(3)} = \frac{0.3}{0.6} = 0.5 \text{ Ans}$$

$$P(Y=3/X=3) = \frac{P(3,3)}{P(3)} = \frac{0.15}{0.6} = 0.25 \text{ Ans}$$

$$\begin{aligned} \text{iii) } F(2,2) &= P(X \leq 2, Y \leq 2) \\ &= P(2,1) + P(2,2) \\ &= 0.1 + 0.2 \\ &= 0.3 \text{ Ans} \end{aligned}$$

$$\begin{aligned} \text{i) } F(3,2) &= P(X \leq 3, Y \leq 2) \\ &= \cancel{(3,1)} + \cancel{(3,2)} + \cancel{(2,1)} + \cancel{(2,2)} + 0 \\ &= P(3,1) + P(3,2) + P(2,1) + P(2,2) \\ &= 0.15 + 0.3 + 0.1 + 0.2 \\ &= 0.75 \text{ Ans} \end{aligned}$$

iv)

$$V(X) = E(X^2) - [E(X)]^2$$

$$\begin{aligned} &= x_1^2 P(x_1) + \cancel{x_1^2 P(x_1)} + x_2^2 P(x_2) - [x_1 P(x_1) + \cancel{x_1 P(x_1)} + x_2 P(x_2)]^2 \\ &= 2^2 P(2) + 3^2 P(3) - [2P(2) + 3P(3)]^2 \\ &= 4 \times 0.4 + 9 \times 0.6 - (0.4 + 0.6)^2 \end{aligned}$$

$$\therefore V(X) = 0.24 \text{ Ans}$$

$$V(Y) = E(Y^2) - [E(Y)]^2$$

$$= \cancel{y_1^2} P(y_1) + y_2^2 P(y_2) + y_3^2 P(y_3) - [y_1 P(y_1) + y_2 P(y_2) + y_3 P(y_3)]^2$$

$$= 1^2 \times 0.25 + 2^2 \times 0.5 + 3^2 \times 0.25 - (1 \times 0.25 + 2 \times 0.5 + 3 \times 0.25)^2$$

$$\therefore V(Y) = 0.5 \text{ Ans.}$$

$$\Rightarrow P(X=3; Y=2) = 0.3$$

$$P(X=3) \times P(Y=2) = 0.6 \times 0.5 = 0.3$$

$$\therefore P(X=3, Y=2) = P(X=3) \cdot P(Y=2) = 0.3$$

Hence X and Y are independent.